



# Extract variables

One feature per block of the tree that turns what its features produced into named variables and query variables -- regions, cuts, chain ends, the largest faces -- instead of a query variable per name.

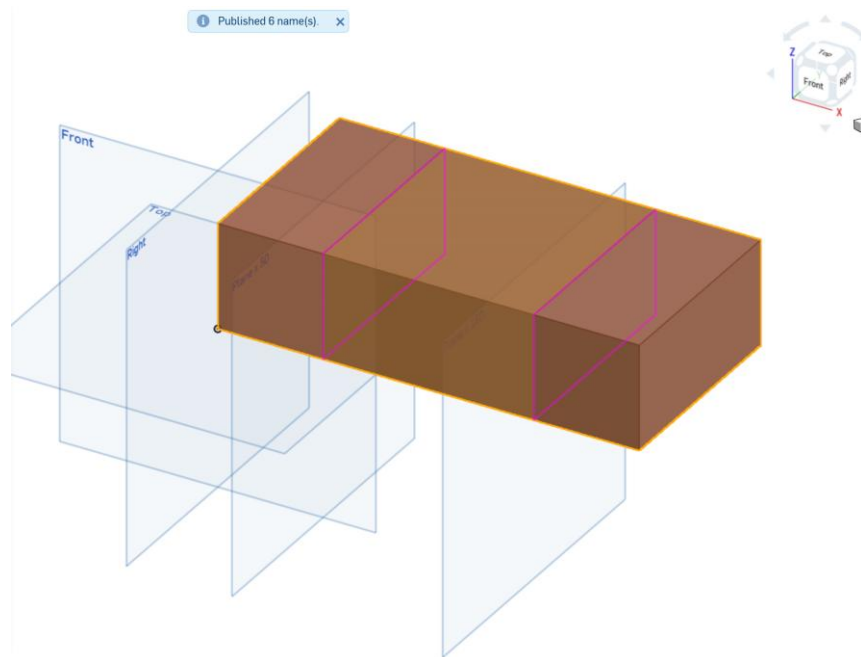


## What it does

A variable-driven model needs dozens of named selections so later features keep working when the model changes. Native query variables give one node per name, picked by hand, and cannot express a region, a chain end or 'the two largest faces'. Extract variables publishes them all from one node, from what the source features published.

### Use it when

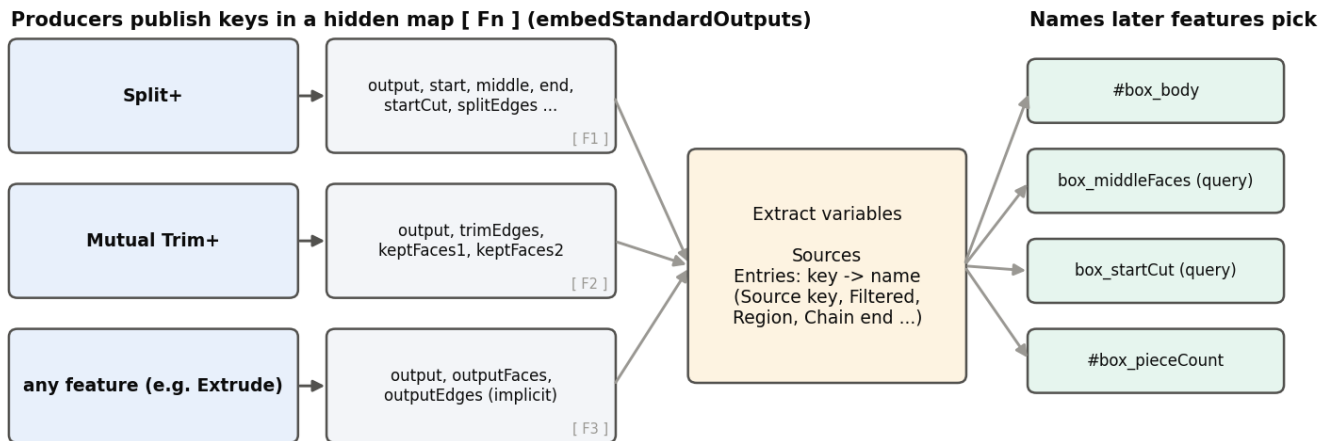
- A block of features feeds many later selections
- A selection must be computed (region, chain end, largest N)
- Names must survive changes upstream



*Examples studio: Show highlights what the entries publish; 'Published 6 name(s).'*



# Producers and the consumer



## Producers

Our features publish keys (output, regions, cuts, ends ...) in a hidden map; built-ins still offer output and modified\* keys.

## Addresses

A key offered by one source is its bare name; by several, key@1, key@2 ... (Print keys lists them).

## Always present

A producer's keys exist even when empty, so an entry never breaks after an edit.



# Hold, Track, Evaluate on use

Extract variables publishes  
edge E (at x 100)



Move boundary extends  
E by 10 mm (to x 110)

a later feature uses the name

## Hold (default)

what was there at extraction, followed through edits that keep it

-> the extended edge

## Track

also whatever a later split or rebuild makes of it

-> the extended edge (+ its pieces)

## Evaluate on use

the raw query, re-resolved wherever the name is used

-> whatever it finds there

Verified by the Tracking tests studio (4/4): Hold already follows the extend; modifiedEdges of the Move boundary gives the moved edge + its 2 lengthened neighbours.

### Hold (default)

What was there, followed through edits that keep it -- verified: follows a 10 mm Move boundary extend.

### Track

Also what later splits and rebuilds make of it.

### Evaluate on use

Re-resolved wherever used: the most live, and the easiest to get wrong.



# The dialog

1

2

3

4

6

Extract variables: box ba... ✓ ✗

Sources  
Box ✕  
Split faces into bands ✕

Prefix box

Add keys from sources

Entries ↑↓ CLEAR ✕

body output@1 ✕

Source key ▼

Source key output@1

Published name body

☒ Show

☐ Evaluate on use

☐ Track

largestFaces outputFaces... ✕

Filtered ▼

Source key outputFaces@1

Published name largestFa...

☐ Show

Faces ▼

Any ▼

Body name contains

☒ Keep only the largest

How many 2

☐ Track

edgesAtCorner outputEdg... ✕

Closest to point ▼

Source key outputEdges@1

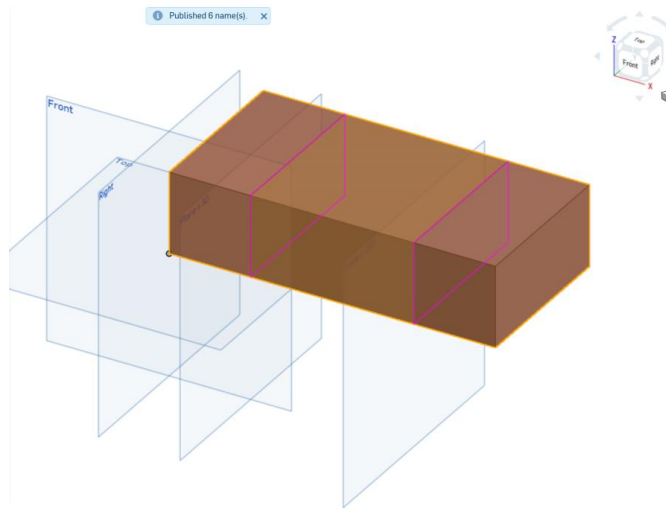
☐ Print keys

| # | Parameter             | What it means                                                                                                                     |
|---|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1 | Sources               | The features to read (sorted by tree order).                                                                                      |
| 2 | Prefix                | Joined to every name with _ (box -> #box_body).                                                                                   |
| 3 | Add keys from sources | Button: one entry per address not yet listed.                                                                                     |
| 4 | Entries               | Per name: Type, Source key, Published name, Show, Track / Evaluate on use; Filtered adds entity type, body type, name, largest N. |
| 5 | Print keys            | List every address with its description.                                                                                          |
| 6 | Manifest              | Optional: one map variable listing everything published.                                                                          |

Entry types: Source key, Filtered, Closest to point, Shared edges, Chain end, Edges between points, Bridging curve input, Region.



## Example: box bands



### Sources: a Box and a split of its faces into bands; prefix 'box'

Entries publish the body (output@1), the two largest faces (Filtered, keep 2), the edges at a corner (Closest to point), shared edges and two regions -- 6 names, plus a manifest. With Show on, the published body and cuts are highlighted.



# Outputs and messages

## Published for Extract variables

| Key                        | What it holds                                                               |
|----------------------------|-----------------------------------------------------------------------------|
| #<prefix>_<name>           | A value entry: an ordinary variable.                                        |
| <prefix>_<name><br>(query) | A query entry: a native query variable, in the dropdowns of later features. |
| manifest (optional)        | One map variable listing everything published.                              |

## Messages

|         | Message / what to do                                                                                                                         |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Info    | Published N name(s).<br>Empty names are published too, and listed.                                                                           |
| Info    | Nothing published yet: press Add keys...                                                                                                     |
| Warning | Entry k (key): reason.<br>Not published: unknown or ambiguous key (choices listed), a value where geometry was expected, a missing point ... |
| Error   | a name used twice<br>Two entries publish the same name.                                                                                      |



## Tips and limits



### **Press Add keys, then prune**

It lists everything the sources offer; delete what you do not need and rename the rest.



### **Prefer Hold**

Use Track for things later split; Evaluate on use only when you know why.



### **Adding a source**

Every source offers output: a second source makes the bare 'output' ambiguous-- use output@1.



## Where it lives

| <b>Document</b>  | Variable_tools (a47f90bfa6b17a59e20cebd0), V2                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Tabs</b>      | extract_variables, extract_variables_utils, extract_outputs (the producer library, pinned by every producer)     |
| <b>Tests</b>     | Tracking tests studio (4/4), variable_tools_tests (Region, 6/6)                                                  |
| <b>Examples</b>  | Examples studio: Extract variables: box bands                                                                    |
| <b>Explainer</b> | docs/explainers/variable_tools/variable_tools_explained.md                                                       |
| <b>Related</b>   | Every producer: Split+, Mutual Trim+, Offset+, Thicken+, Orient to reference, Clean wire, Map / Merge curve, ... |